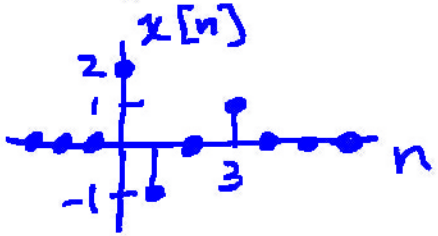
Impulse Response and LTI Systems Examples

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Example: An LTI system has impulse response $h[n] = (1/3)^n u[n]$. Find the output if the input is



Solution: Write $x[n] = 2\delta[n] - \delta[n-1] + \delta[n-3]$

1) Time invariance: $\delta[n-k] \longrightarrow h[n-k]$

so $\delta[n-1] \rightarrow (1/3)^{n-1} u[n-1]$

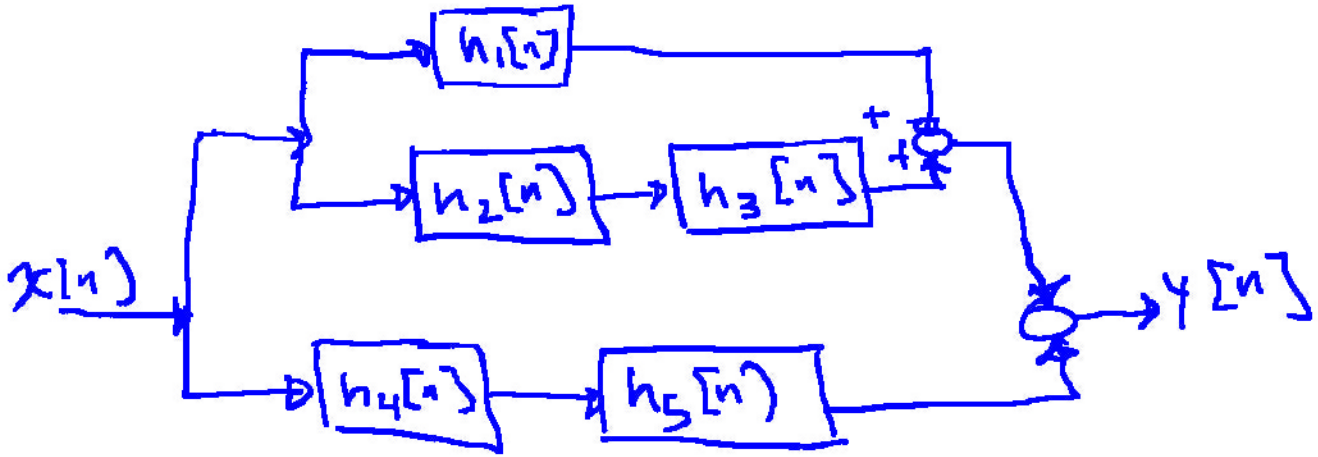
$\delta[n-3] \rightarrow (1/3)^{n-3} u[n-3]$

2) Linearity

$$x[n] = 2\delta[n] - \delta[n-1] + \delta[n-3] \rightarrow y[n] = 2h[n] - h[n-1] + h[n-3]$$

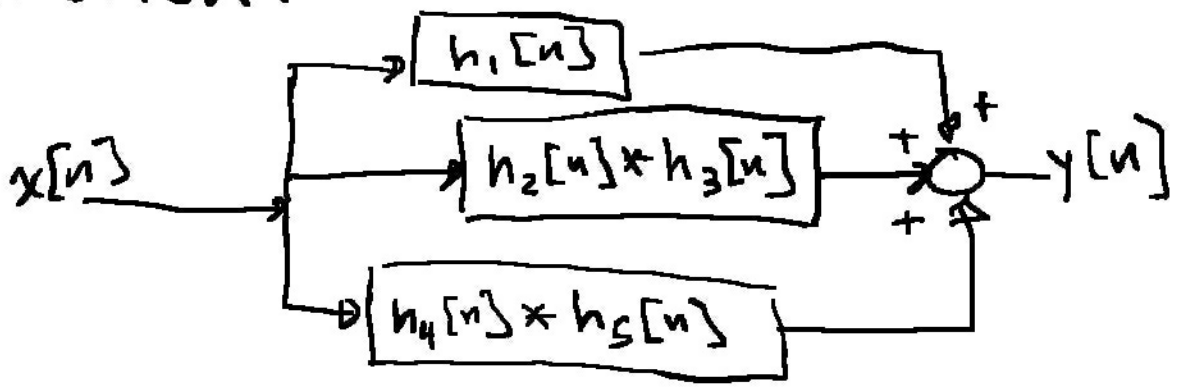
$$= 2(1/3)^n u[n] - (1/3)^{n-1} u[n-1] + (1/3)^{n-3} u[n-3]$$

Example: Consider the interconnection of LTI systems shown. Find the impulse response of the equivalent system.



$$h[n] = h_1[n] + h_2[n] * h_3[n] + h_4[n] * h_5[n]$$

Solution:



$$= h_1[n] + h_3[n] * h_2[n] + h_5[n] * h_4[n]$$

Example: A system has impulse response $h[n] = (-1/2)^n u[n]$.

Is this system stable?

Solution: Is $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$?

$$\sum_{n=-\infty}^{\infty} |(-1/2)^n| u[n] = \sum_{n=0}^{\infty} 1/2^n = \frac{1}{1-1/2} = 2$$

Yes, stable

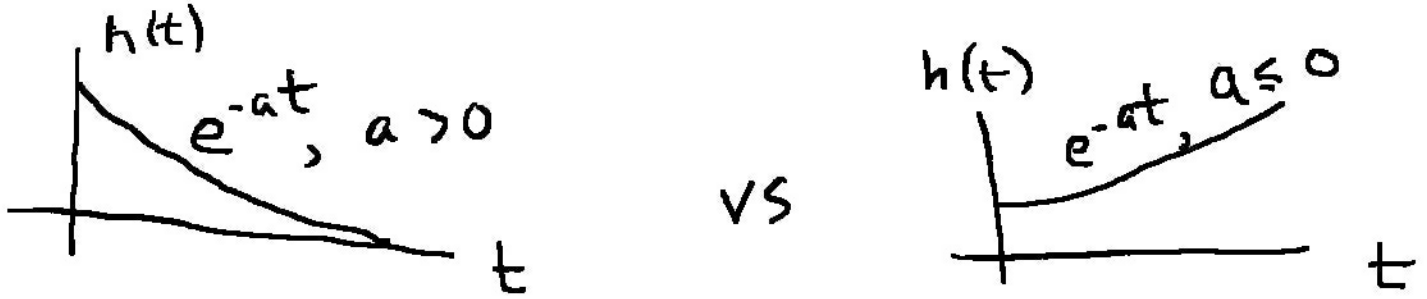
Example: A continuous-time system has impulse response $h(t) = e^{-at}u(t)$. Find the values of a for which it is BIBO stable.

Solution: Is $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$?

$$\int_{-\infty}^{\infty} |e^{-at}|\,u(t)\,dt = \int_{0}^{\infty} e^{-at}\,dt = \left.-\frac{1}{a}\,e^{-at}\right|_{0}^{\infty}$$

$$= \begin{cases} 1/a, & a > 0 \\ \infty, & a \le 0 \end{cases}$$

System is BIBO stable for $a > 0$



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